

Architectural Particular Specification

SOLID SURFACING MATERIAL

SECTION 8.11

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PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. Supply, and install solid polymer fabrication to match reference type, colour, finishes and pattern as indicated on the Drawings and as specified for the following items as shown in Drawing:
 - 1. Countertops, back splashes, end splashes in pantry and
 - 2. Countertops, back splashes, end splashes in lavatories with concealed cistern; and
 - 3. Other area as shown in the Drawings.
- B. Verify adhesion to substrate, sub-frame anchor or similar with adhesion sealant recommended by solid polymer manufacturer.
- C. Protection: Unless as otherwise approved and recommended in writing by the manufacturer, proper protection should be provided for completed works / installations from damage and deterioration from weather/humidity exposure until subsequent facing or finish materials are applied or installed.

1.02 REFERENCE

Comply with all the following relevant standards and in this section. In case of discrepancies or differences between two or more standards / sections, the Contractor shall refer to the Architect for confirmation.

- A. ABS Standards:
 - 1. BS 476-4: Fire Tests on Building Materials and Structures. Non-combustibility test for Materials.
 - 2. BS 476-7: Fire Tests on Building Material and Structures Method of Test for Fire propagation for Products. All works shall be Class 1 or 2 Rate of Surface Spread of Flame as per BS 476-7 or its international equivalent.
 - 3. BS 476-20: Fire tests on building materials and structures. Method for determination of the fire resistance of elements of construction (general principles).
 - 4. BS 476-22: Fire tests on building materials and structures. Methods for determination of the fire resistance of non-load bearing elements of construction.
 - 5. BS 476-23: Fire tests on building materials and structures. Methods for determination of the contribution of components to the fire resistance of a structure.
 - 6. BS 476-24: Fire tests on building materials and structures. Method for determination.
- B. Local Codes:
 - 1. Fire Safety Requirements by FSD
 - 2. Design Manual - Barrier Free Access 2008

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

- A. General: Conform to solid polymer surfacing manufacturer's technical bulletins, drawings, guides, and manuals.
 - 1. Fabricate in precisely controlled shop environment to the maximum extent practical, to sizes and shapes indicated on Drawings and approved shop drawings. Except for jointing or seaming of large pieces or sections, or repair of minor damage incurred after shop fabrication, field fabrication is not acceptable
 - 2. To the maximum extent possible, fabricate adjacent pieces or sections using solid polymer surfacing materials as sequentially manufactured.
- B. The material, structure, installation, workmanship and performance will be in proper and normal function in accordance with the contract requirements for a period of not less than TEN years from the date of substantial completion of the contract works as certified by the Architect.
- C. Joint and seaming:
 - 1. Unless otherwise indicated on the Drawings, fabricate pieces or sections from largest single manufactured material sizes as practical to minimize number of joints.
 - 2. Form joints and seams using specified sealant adhesives. Finish to be inconspicuous and to match adjacent surfaces.
- D. Cutouts: Accurately and precisely cut to templates furnished by manufacturer or fabricator of item to be attached to, installed in or upon, or secured to solid polymer fabrications. Radius all corners of cutouts. Square corner cutouts are not acceptable.
- E. Reinforcement and support: Reinforce or otherwise support cutouts, joints and seams, overhangs, and other similar conditions, as recommended by solid polymer surfacing manufacturer, using solid polymer surfacing materials.
- F. Make provisions for isolating, insulating and expansion/contraction as recommended by solid polymer surfacing manufacturer.
- G. Unless otherwise indicated on Drawings, form backsplashes, endsplashes, and similar internal corners with integral radius coves, without visible seams or use of accessory sealant materials.
- H. For fabrication subject to exterior wind load, such as rooms with open ends, design support and anchor to withstand wind load in accordance with Code of Practice on Wind Effects-Hong Kong, or Factory Mutual (FM) Global 'Wind Uplift and Outward Pressure Design Criteria' whichever is more stringent. FM Global standards are available for inspection and information upon request.
- I. Inlay designs:
 - 1. Liquid:
 - a) Rout, abrasive-blast, etch, or otherwise provide inlay design and patterns as indicated on Drawings.
 - b) Overfill inlay design, ensuring that air bubbles or similar voids do not form.
 - c) Allow to fully cure, and finish inlay to match adjacent surfaces.

2. Metal, wood, ceramic, or other dissimilar materials:
 - a) Rout or otherwise provide for inlay materials.
 - b) Set inlay materials using adhesive, silicone sealant adhesive, or grout as recommended by solid polymer surfacing manufacturer.
- J. Thermoforming:
 1. Fabricate moulds to required shapes. Provide self draining, countertop surface with bull-nosed front edge and coved back profile indicated on drawings and approved shop drawings.
 2. Heat and form solid polymer surfacing materials in molds prior to jointing seaming.
 3. Ensure that solid polymer surfacing materials do not blister, craze, crack, or exhibit similar aesthetic deficiencies on exposed surfaces.
- K. Finishing:
 1. Edges and corners: Rout to uniform radii and finish uniformly smooth to match planar surfaces.
 2. Planar surfaces: Sand with progressively finer microfinishing films and buff/polish to a uniform matte (less than 20 in accord with ASTM D523), semi-gloss (25 to 50 in accord with ASTM D523) or high gloss (greater than 55 in accord with ASTM D523) finish to match samples.
- L. Defective or inaccurate solid polymer fabrications not in conformance with the Drawings details and approved shop drawings, not within specified tolerances, or exhibiting similar deficiencies are not acceptable.
- M. Visible repairs in conspicuous areas of installed work are not acceptable.
- N. Allowable tolerances:
 1. Variation in component size: $\pm 3\text{mm}$ ($\pm 1/8$ in.).
 2. Location of openings: $\pm 3\text{mm}$ ($\pm 1/8$ in.) from indicated location

1.05 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

- A. Product Data:
 1. For each variety of solid surfacing, include data on physical properties required by referenced British Standards.
 2. For solid surfacing accessories and other manufactured products.

3. Proof of properties as appropriate:

- a) Certifications: Solid polymer surfacing manufacturer's or authorized distributor's written certification that fabricator/installer is registered and certified to perform the type of work required of this Project.
- b) Adhesion
- c) Calculation or test certificate on adhesion capability of selected sealant adhesive.

B. Samples for Selection:

- 1. For each variety of solid surfacing, in sets of samples with minimum 300 mm square. Include three or more samples in each set and show the full range of variations in appearance characteristics expected in completed Work. Samples sets shall establish the standard by which solid surfacing shall be judged.
- 2. Joint materials involving color selection.

C. Sealant Compatibility Test Report: From sealant manufacturer, complying with requirements in Division 07 Section "Joint Sealants" and indicating that sealants will not stain or damage solid surfacing.

D. Maintenance data and instructions

E. Solid surfacing manufacturer's care and maintenance video and data, including cleaning and repair instructions. Repair instructions must include specific and descriptive narrative of finishing materials and processes/methods utilized for specific types of finishes for the work of this Project.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

- A. Deliver solid surfacing units to Project site in manufacturer's original, labeled packaging, in undamaged condition.
- B. Store and handle solid surfacing units off the ground, under cover, and in a dry location to prevent their deterioration or damage due to moisture, temperature changes, contaminants, corrosion, and other causes. If units become wet, do not place until units are in an air-dried condition.
- C. Solid surfacing units too large to lift by hand shall be lifted with wide-belt slings; do not use wire rope or ropes that might cause staining. Move solid surfacing, if required, using dollies with cushioned wood supports.
- D. Store solid surfacing on pallets with non-staining separators and non-staining, waterproof covers. Ventilate under covers to prevent condensation.
- E. Protect materials from freezing and overheating in accordance with manufacturer's instructions.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

(NOT USED)

1.12 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER/SUPPLIERS

- A. Subject to compliance with the Project Contract Documents, provide tiles, paves and required accessories from one or more of the following approved manufacturers/suppliers, or equivalent approved by the Architect:
1. Speed Top (Hong Kong) Ltd.
DuPont® Corian® Hong Kong/Macau Authorized Distributor
Tel: 2435 4427
Fax: 2431 4165

2.02 MATERIALS

- A. The solid surfacing material and accessories shall be non-combustible to comply with the requirement of fire certificates, anti-scratch, anti-stain, and durable.
- B. Solid polymer surfacing: Chemical- and stain- resistant, homogeneous, filled resin conforming to ANSI Z124.3, with Class A rating in accord with ASTM E84. Except as required by design or recommended by solid polymer surfacing manufacturer, provide 13mm (1/2 in.) thickness. Colour, pattern and finish shall be as indicated on Drawings or as selected or approved by Architect. Shall be manufactured to the following standards:
1. Surface resistance to cracks, chips and blisters.
 2. Subsurface consistency to verify that there are no air voids larger than 1/16" below the gel-coat surface.
 3. Load testing to determine resistance to breakage under heavy weight.
 4. Stain resistance testing.
 5. Cleanability test to assure surface hardness.
 6. Thermal shock testing to assure resistance to damage from hot and cold water exposure.
- B. Adhesive, Sealant, And Solid Surfacing Accessories
1. Joint/seam adhesive: Solid polymer surfacing manufacturer's standard, chemically bonding joint/seam adhesive. Colour to match solid polymer surfacing.
 2. Sealant adhesive: Mildew-resistant, silicone sealant adhesive. Except as indicated otherwise, colour shall match or complement solid polymer surfacing at exposed locations.
 3. Inlay materials:
 - a) Liquid: Solid polymer surfacing manufacturer's standard liquid inlay materials. Unless otherwise indicated on Drawings or approved shop drawings, colour as selected by Architect.
 - b) Metal, wood, ceramic or other dissimilar materials: Except as otherwise indicated on the Drawings, as selected by Architect.
 4. Bowl (Sink) Mounting Hardware: Approved bowl clips, brass panel inserts, and fasteners recommended by solid polymer and bowl (sink) manufacturer for attachment of under mount bowls (sinks).

C. Solid Surfacing Countertops

1. Nominal Thickness: Provide minimum 10mm thick solid surfacing slab, unless otherwise indicated on Drawings. Gage backs to provide units of identical thickness.
2. Edge Detail: Straight, with 5mm radius bullnose on horizontal surface unless otherwise indicated on Drawings.
3. Splashes: Provide 10mm thick backsplashes and end splashes, unless otherwise indicated on Drawings.
 - a) Height: 100 mm, unless otherwise indicated on Drawings.
 - b) Top-Edge Detail: Straight, slightly eased at top.
4. Joints:
5. Fabricate countertops less than 3m long without joints.
6. Fabricate countertops more than 3m long in sections for joining in field, with joints at locations as follows, unless otherwise indicated on Drawings:
 - a) Bonded Joints Width: Maximum 0.8mm.
 - b) Splined Joints: Accurately cut kerfs in edges at joints for insertion of metal splines to maintain alignment of surfaces at joints adjacent to cutouts. Make width of cuts slightly more than thickness of splines to provide snug fit.
7. Form joints and seams using specified solid surface sealant adhesives. Finish to be inconspicuous and to match adjacent surfaces.
8. Cutouts and Holes:
 - a) Under-counter Plumbing Fixtures: Make cutouts for plumbing fixtures in shop using template or pattern furnished by plumbing fixture manufacturer. Form cutouts to smooth, even curves, to profile as indicated on Drawings.
 - b) Above-counter Plumbing Fixtures: Make cutouts for plumbing fixtures in shop using template or pattern furnished by plumbing fixture manufacturer. Form cutouts to smooth, even curves.
 - c) Provide vertical edges, slightly eased at juncture of cutout edges with top and bottom surfaces of countertop and projecting 5mm into plumbing fixtures opening.
 - d) Fittings: Drill countertops in shop for plumbing fixtures, under-counter soap dispensers, and similar countertop accommodated items.
9. Thermoforming:
 - a) Fabricate molds to required shapes. Provide self-draining, countertop surface with bullnosed front edge and coved back profile indicated on drawings and approved shop drawings.
 - b) Heat and form solid surfacing materials in molds prior to jointing seaming.
 - c) Ensure that solid surfacing materials do not blister, craze, crack or exhibit similar aesthetic deficiencies on exposed surfaces.
10. Finishing: Edges and corners: Rout to uniform radii and finish uniformly smooth to match planar surfaces.
11. Concealed Fixing:
 - a) Solid surface cap should be applied to cover the fixing area.
 - b) Adhesive with same color has to be applied and polish the surface to seamless

PART 3 EXECUTION

3.01 EXAMINATION

Examine surfaces and conditions indicated to receive solid surfacing, with installer present, for compliance with requirements of Sections in which substrates and related work are specified and for other conditions affecting performance.

- A. For the record, prepare written report, endorsed by installer, listing conditions detrimental to performance of solid surfacing surface.
- B. Proceed with installation only after unsatisfactory conditions have been corrected.
- C. Confirm that prerequisite work has been completed.

3.02 PREPARATION

- A. Advise installers of other Work about specific requirements for placement of inserts and similar items to be used by solid surfacing countertop installer for anchoring solid surfacing countertops. Furnish installers of other work with Drawings or templates showing locations of these items.
- B. Clean dirty or stained solid surfacing surfaces by removing soil, stains, and foreign materials before setting. Clean solid surfacing by thoroughly scrubbing with fiber brushes and then drenching with clear water. Use only mild cleaning compounds that contain no caustic or harsh materials or abrasives. Allow solid surfacing to dry before installing.

3.03 INSTALLATION

- A. Countertops
 - 1. General: Install countertops by adhering to supports with water-cleanable epoxy adhesive.
 - 2. Do not cut solid surfacing in field, unless otherwise indicated on Drawings. If solid surfacing countertops or splashes require additional fabrication specified not to be performed at Project site, return to fabrication shop for adjustment.
 - 3. Set solid surfacing to comply with requirements indicated on Drawings. Shim and adjust solid surfacing to locations indicated on Drawings, with uniform joints of widths indicated and with edges and faces aligned according to established relationships and indicated tolerances. Install anchors and other attachments indicated or necessary to secure solid surfacing countertops in place.
 - 4. Bond joints with solid surfacing adhesive and draw tight as countertops are set. Mask areas of countertops adjacent to joints to prevent adhesive smears.
 - 5. Install metal splines in kerfs in solid surfacing edges at joints adjacent to cutouts and where indicated on Drawings. Fill kerfs with solid surfacing adhesive before inserting splines and remove excess immediately after adjoining units are drawn into position.
 - 6. Clamp units to temporary bracing, supports, or each other to ensure that countertops are properly aligned and joints are of specified width.
 - 7. Miter vertical joints as indicated on Drawings.

8. Install backsplash and end splash by adhering to countertops with solid surfacing adhesive. Mask areas of countertops and splashes adjacent to joints to prevent adhesive smears. Leave 1.5mm gap between splash and wall for filling with sealant. Use temporary shims to ensure uniform spacing.
9. Apply sealant to gaps specified for filling with sealant; comply with Section "Joint Sealants". Remove temporary shims before applying sealant.

3.04 TOLERANCE

- A. Variation from Plumb: For vertical lines and surfaces, maximum 1.5mm in 1200mm.
- B. Variation from Level: Do not exceed 3mm in 3000mm, maximum 6mm.
- C. Variation in Joint Width: Do not vary joint thickness more than $\frac{1}{4}$ of nominal joint width.
- D. Variation in Plane at Joints (Lipping): Do not exceed 0.4mm difference between planes of adjacent units.
- E. Variation in Line of Edge at Joints (Lipping): Do not exceed 0.4mm difference between edges of adjacent units, where edge line continues across joint.

3.05 CLEANING

- A. In-Progress Cleaning: Clean countertops as work progresses. Remove adhesive, grout, mortar, and sealant smears immediately.
- B. Remove and replace solid surfacing countertops of the following description:
 1. Broken, chipped, stained, or otherwise damaged solid surfacing.
 2. Defective countertops and/or defective joints, including misaligned joints.
 3. Countertops and joints not matching selected samples.
 4. Countertops not complying with other requirements indicated on Drawings.
- C. Replace in a manner that results in solid surfacing countertops matching selected samples, complying with other requirements, and showing no evidence of replacement.
- D. Clean solid surfacing countertops within 6 day after completion of sealant installation, using clean water and soft rags. Do not use wire brushes, acid-type cleaning agents, cleaning compounds with caustic or harsh fillers, or other materials or methods that may damage solid surfacing.

3.06 PROTECTION

Protect the material during the construction period so that they will be without any indication of deterioration or damage prior handover.

END OF SECTION